ASYNCHRONOUS ACTIVITY NO. 3

Name It, Draw It

Two-Way IUPAC Nomenclature of Alkanes, Cycloalkanes and Haloalkanes

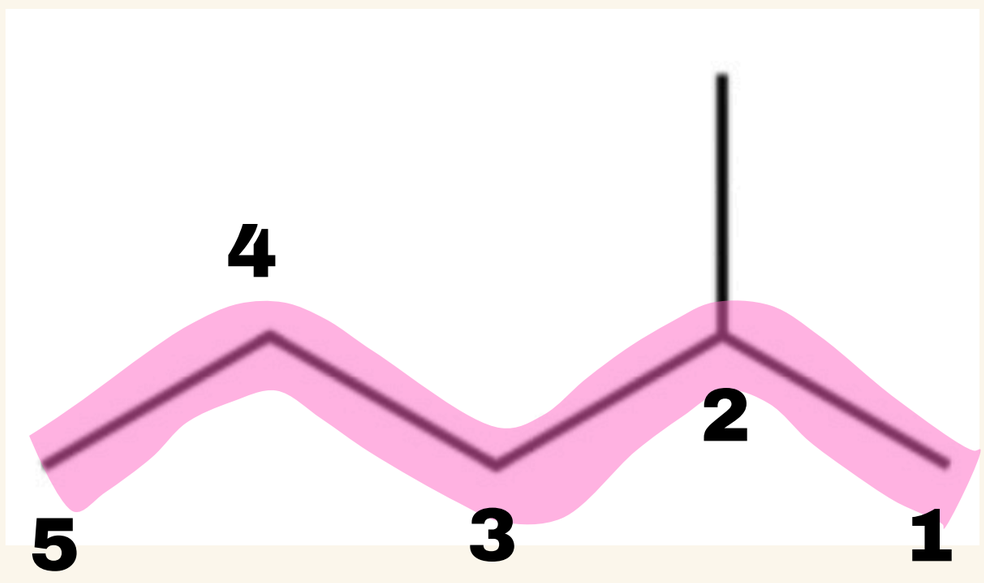
Angel Stefanie C. Gutierrez
BSChE-3B

Part A : STRUCTURE → NAME

Card A01 FOUNDATION

Molecular Formula : C_6H_{14}

1. NUMBERED PARENT



2, NAMING TRAIL

Parent: Longest chain = 5 C → pentane.

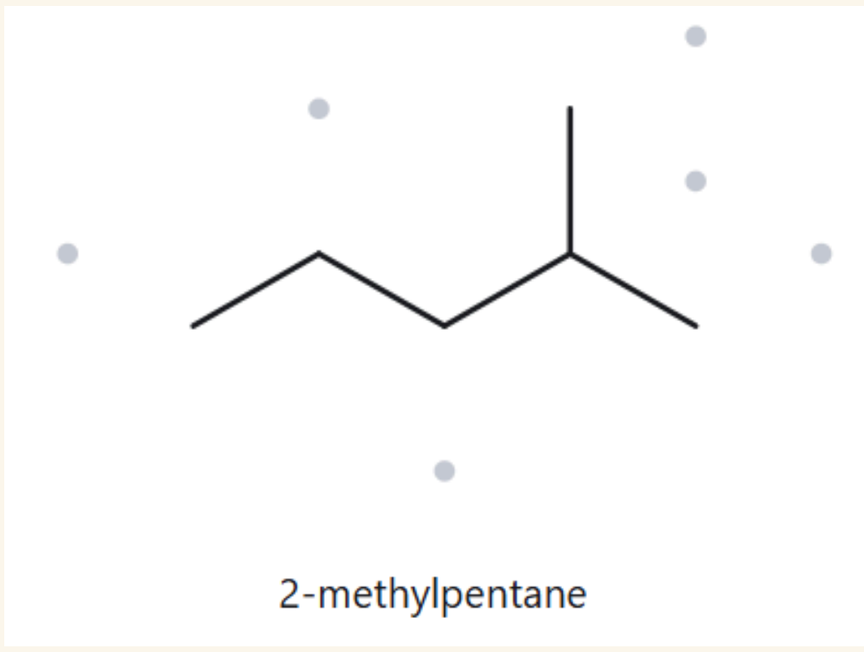
Numbering: Left → right gives 4-methyl; right → left gives 2-methyl. The lower locant is 2, so number from the right.

Substituents: methyl on C_2

3. MY NAME

2-methylpentane

4. THE ALKANE CHECK



Returned name: 2-methylpentane

✓ **MATCH**

VERIFICATION STRIP

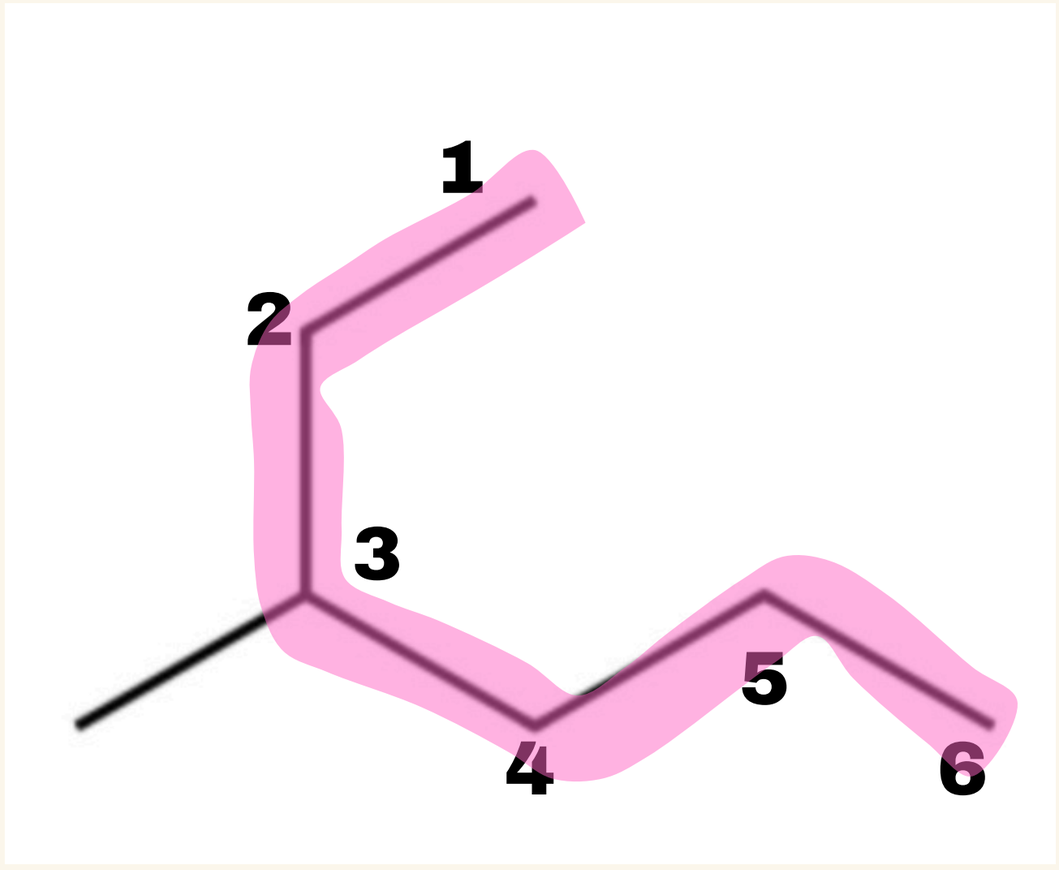
✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part A : STRUCTURE → NAME

Card A02 FOUNDATION

Molecular Formula : C_7H_{16}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 6 C → hexane.

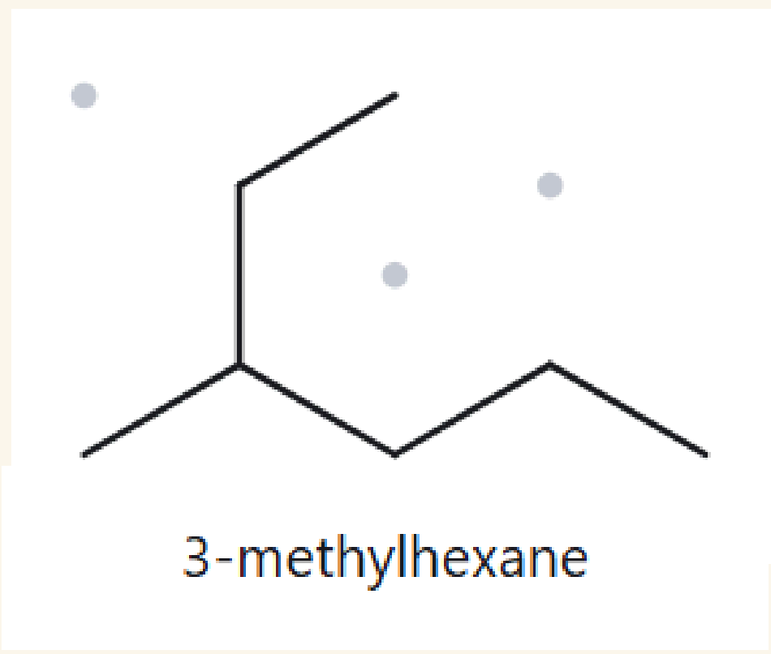
Numbering: Left → right gives 3-methyl; right → left gives 4-methyl. The lower locant is 3, so number from the left.

Substituents: methyl on C_3

3. MY NAME

3-methylhexane

4. THE ALKANE CHECK



Returned name: 3-methylhexane

✓ **MATCH**

VERIFICATION STRIP

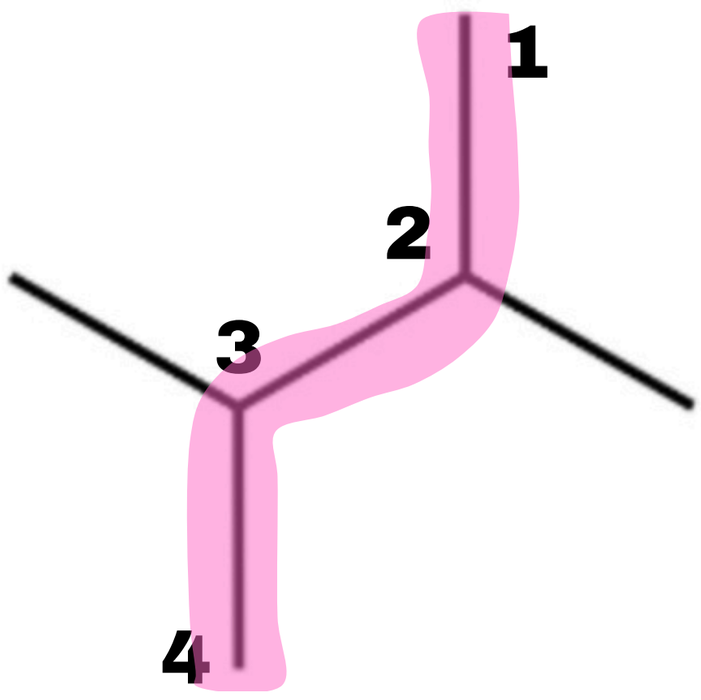
✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part A : STRUCTURE → NAME

Card A03 FOUNDATION

Molecular Formula : C_6H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 4 C → butane.

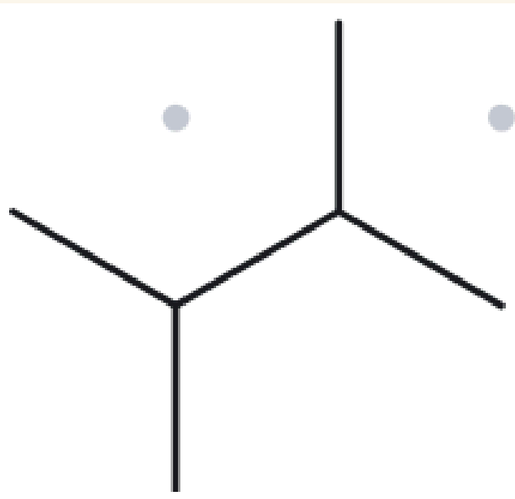
Numbering: Both directions give the same locant set: 2,3

Substituents: methyl on C2 and methyl on C3 → 2,3-dimethyl

3. MY NAME

2,3-dimethylbutane

4. THE ALKANE CHECK



2,3-dimethylbutane

Returned name: 2,3-dimethylbutane

✓ **MATCH**

VERIFICATION STRIP

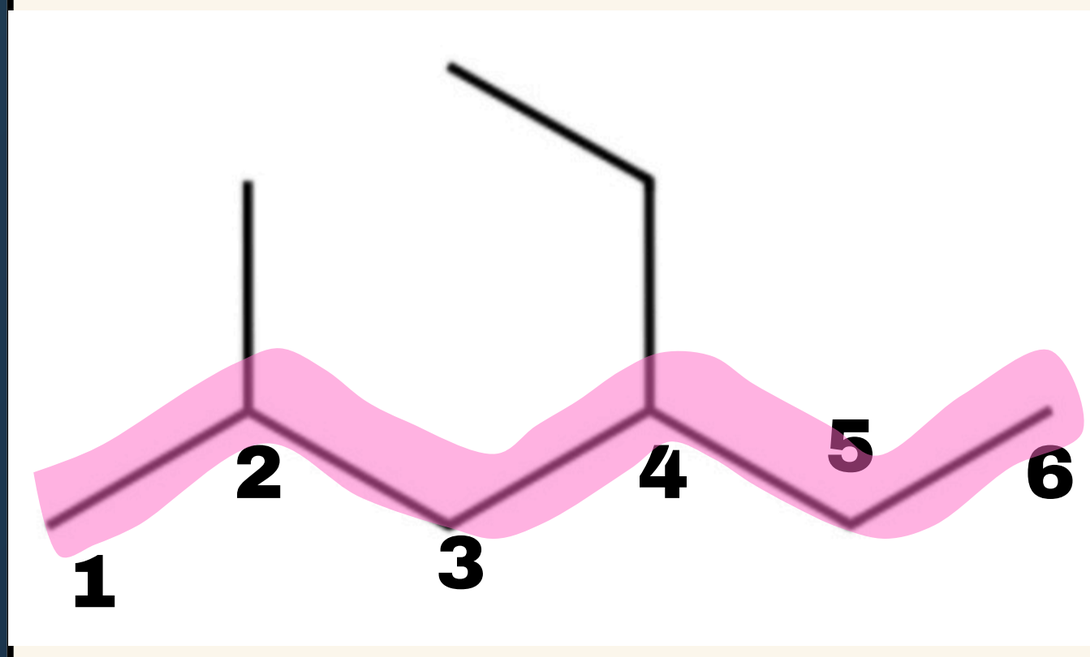
✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

Part A : STRUCTURE → NAME

Card A04 DEVELOPING

Molecular Formula : C_9H_{20}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 6 C → hexane.

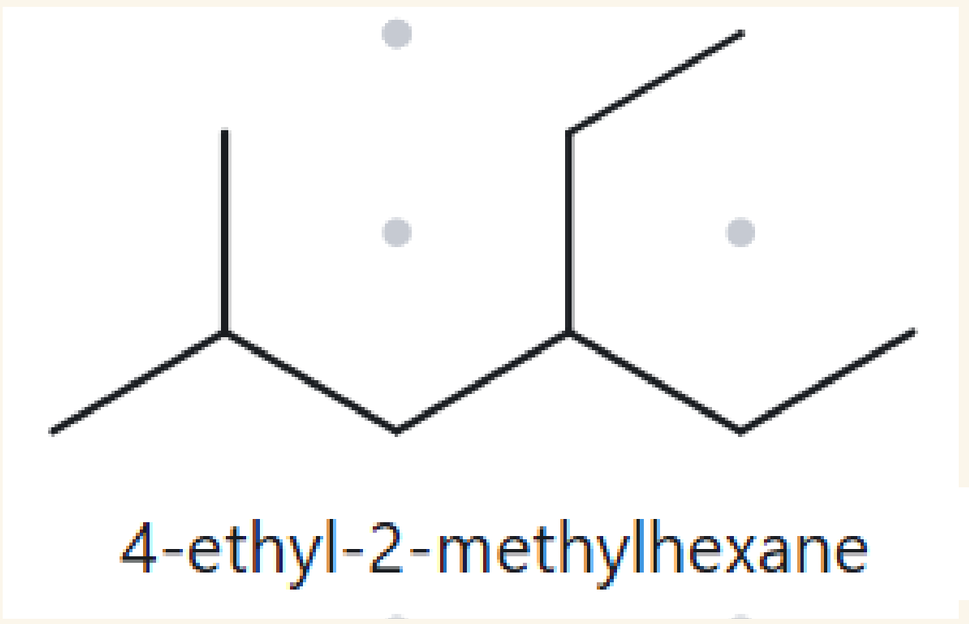
Numbering: Left → right gives 2,4; right → left gives 3,5. The lower locant is 2, so number from the left.

Substituents: methyl on C2 and ethyl on C4 → 4-ethyl-2-methyl

3. MY NAME

4-ethyl-2-methylhexane

4. THE ALKANE CHECK



4-ethyl-2-methylhexane

Returned name: 4-ethyl-2-methylhexane

✓ **MATCH**

VERIFICATION STRIP

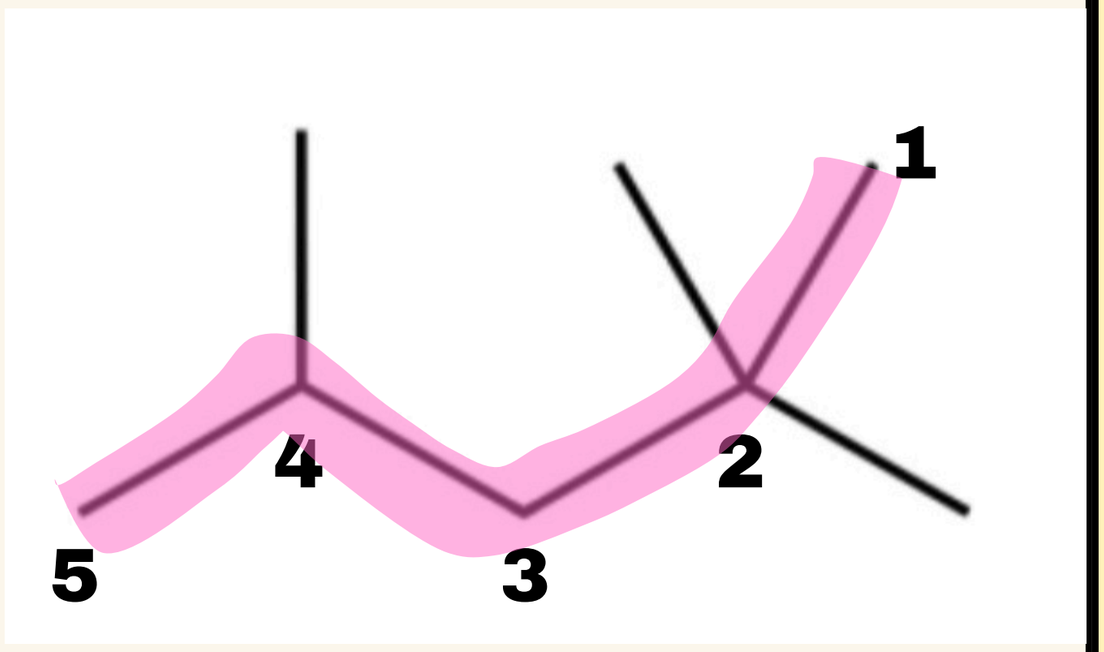
✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part A : STRUCTURE → NAME

Card A05 DEVELOPING

Molecular Formula : C_8H_{18}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 5 C → pentane.

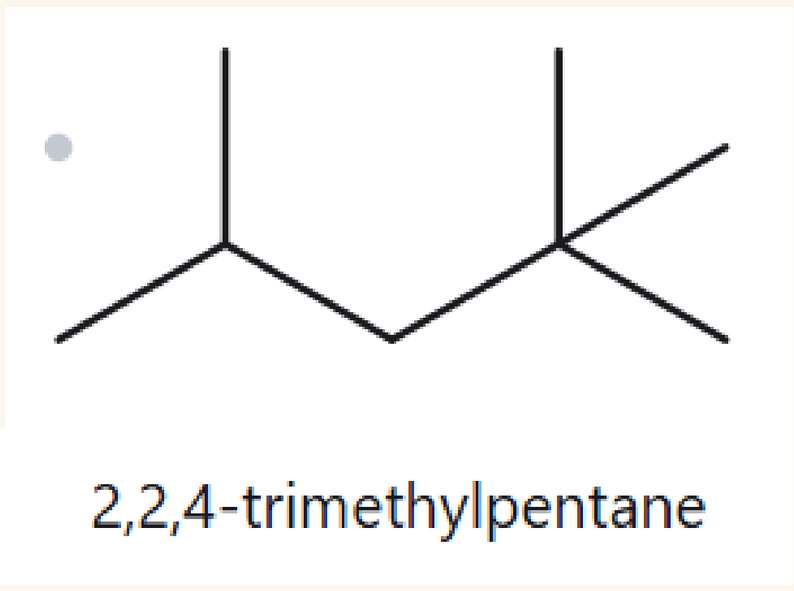
Numbering: Left → right gives 2,2,4; right → left gives 2,4,4. The first point of difference keeps 2,2,4.

Substituents: methyl on C2, C2 and C4 → 2,2,4-trimethyl

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK



2,2,4-trimethylpentane

Returned name: 2,2,4-trimethylpentane

✓ **MATCH**

VERIFICATION STRIP

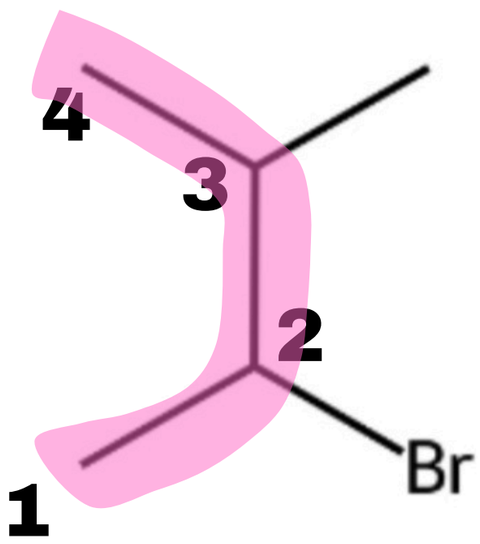
✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part A : STRUCTURE → NAME

Card A06 DEVELOPING

Molecular Formula : $C_5H_{11}Br$

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 4 C → butane.

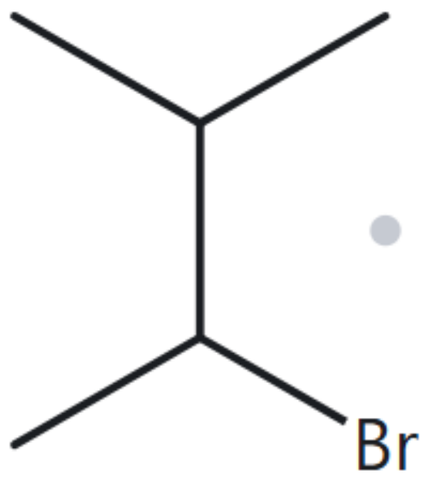
Numbering: Both directions give locants 2 and 3.

Substituents: bromo on C2; methyl on C3

3. MY NAME

2-bromo-3-methylbutane

4. THE ALKANE CHECK



2-bromo-3-methylbutane

Returned name: 2-bromo-3-methylbutane

✓ **MATCH**

VERIFICATION STRIP

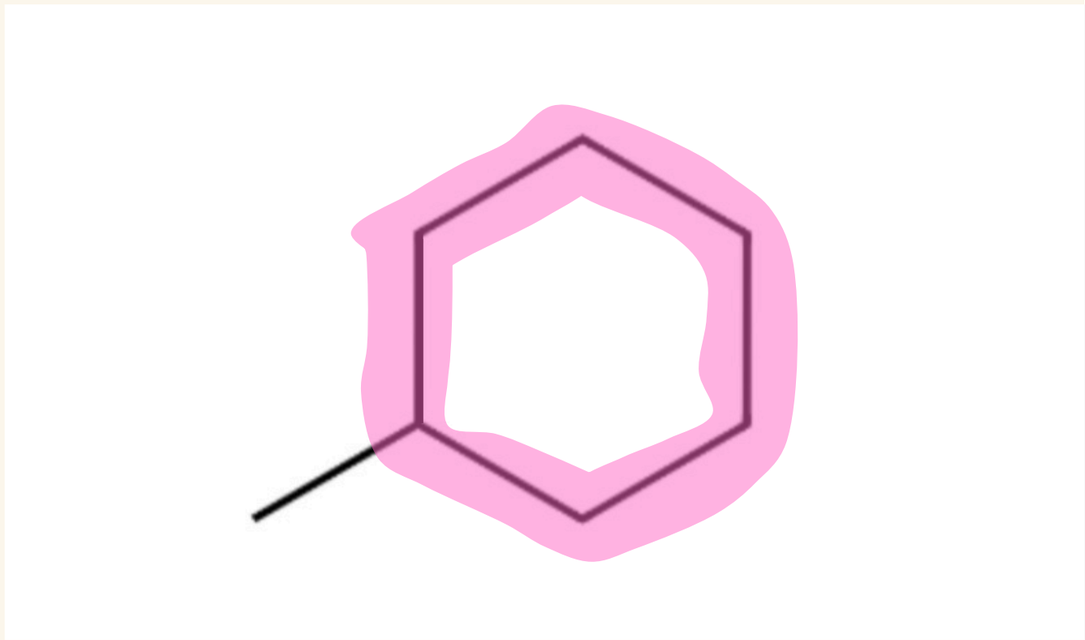
✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part A : STRUCTURE → NAME

Card A07 DEVELOPING

Molecular Formula : C_7H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: The ring has 6 C → cyclohexane.

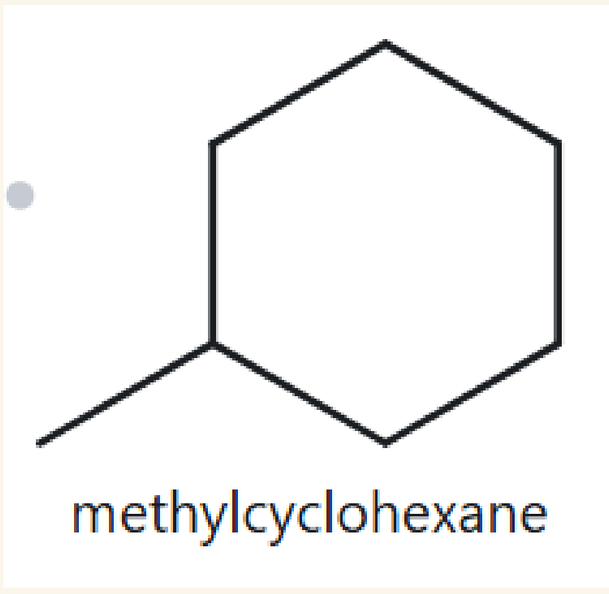
Numbering: Only one substituent is present, so a locant is unnecessary.

Substituents: methyl

3. MY NAME

methylcyclohexane

4. THE ALKANE CHECK



Returned name: methylcyclohexane

✓ **MATCH**

VERIFICATION STRIP

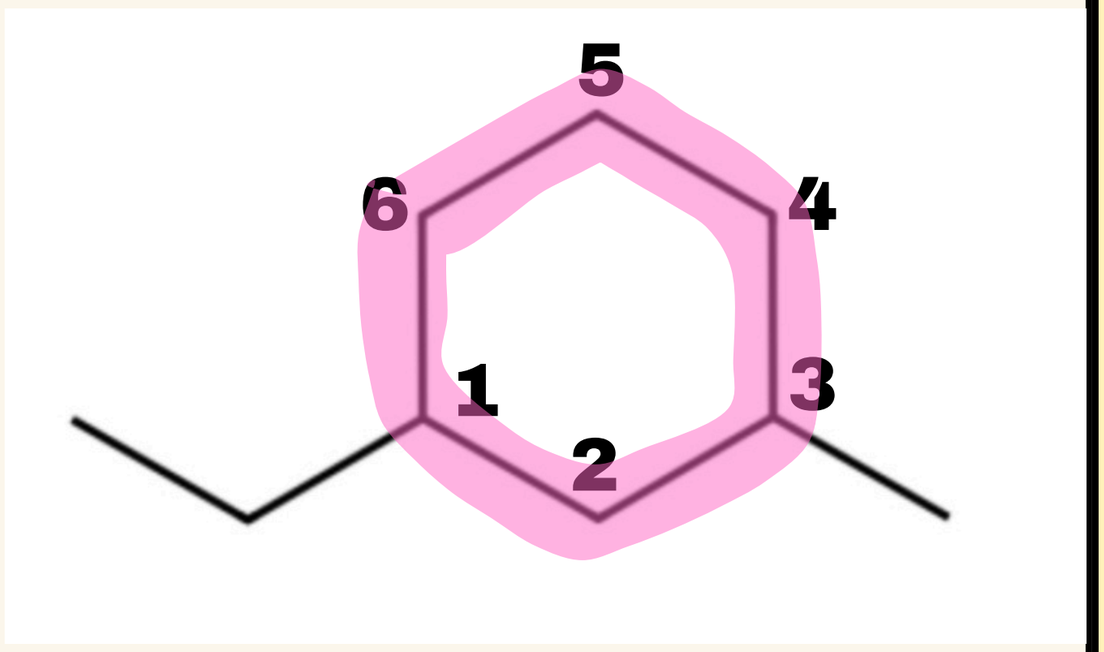
✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part A : STRUCTURE → NAME

Card A08 CHALLENGE

Molecular Formula : C_9H_{18}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: The ring has 6 C → cyclohexane.

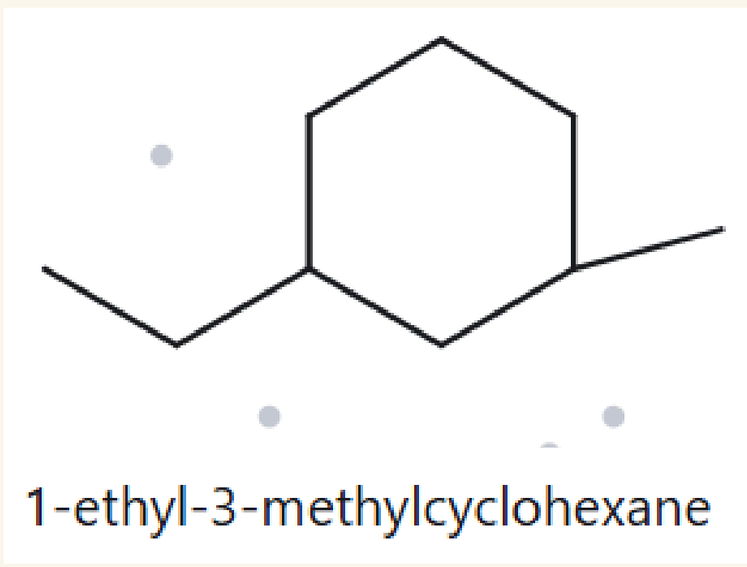
Numbering: The substituents can be numbered 1,3 either way.

Substituents: ethyl on C1; methyl on C3

3. MY NAME

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK



Returned name: 1-ethyl-3-methylcyclohexane

✓ **MATCH**

VERIFICATION STRIP

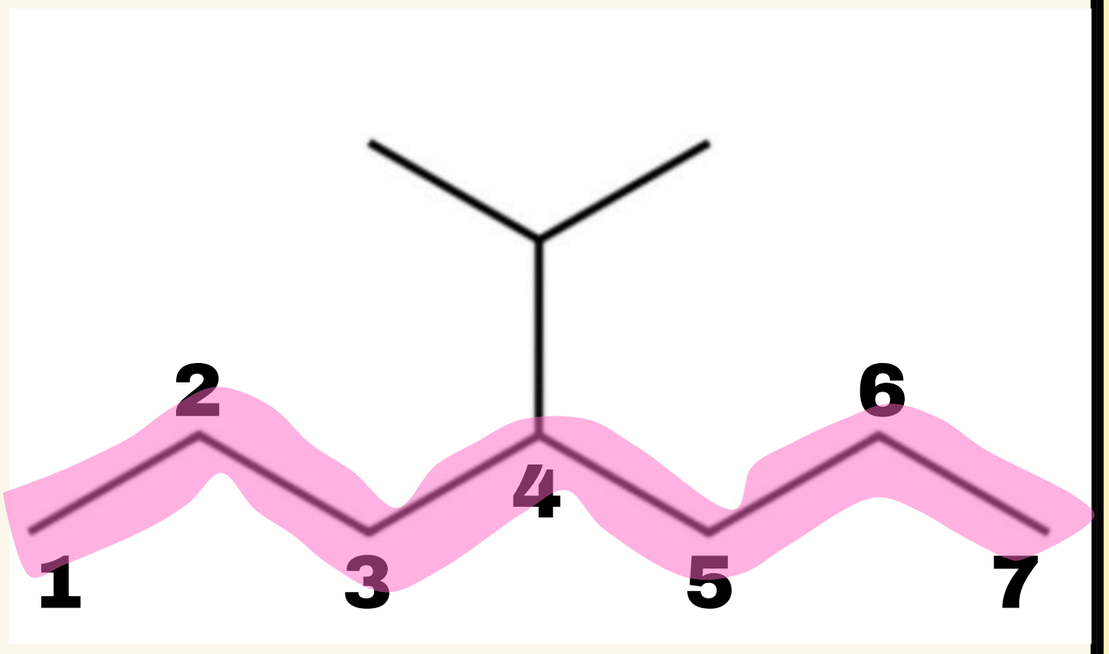
✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part A : STRUCTURE → NAME

Card A09 CHALLENGE

Molecular Formula : $C_{10}H_{22}$

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 7 C → heptane.

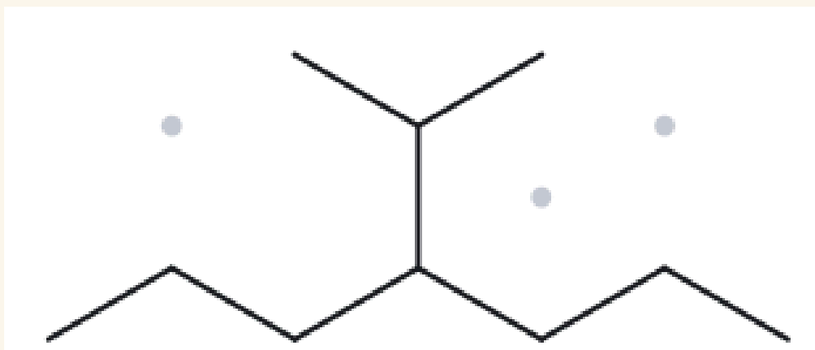
Numbering: Left → right and right → left both give locant 4.

Substituents: propan-2-yl on C4.

3. MY NAME

4-(propan-2-yl)heptane

4. THE ALKANE CHECK



Returned name: 2-methyl-3-propylhexane

NOT MATCH

The longest continuous chain is 7 carbons (heptane), not 6. The upper branch is an isopropyl group, not separate methyl and propyl substituents.

VERIFICATION STRIP

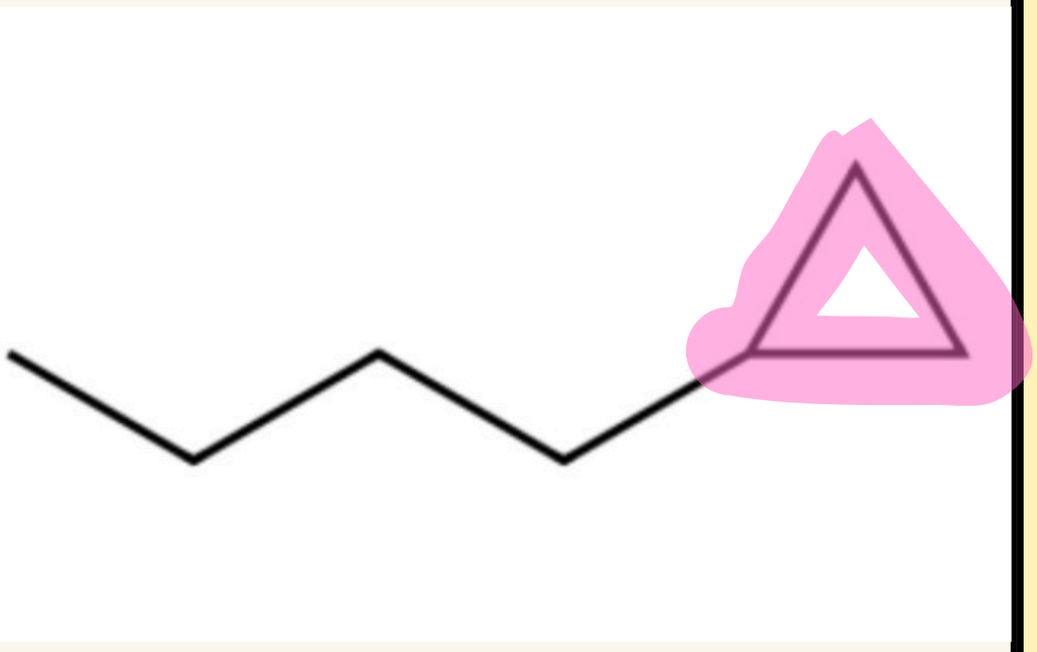
✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part A : STRUCTURE → NAME

Card A10 CHALLENGE

Molecular Formula : C_8H_{16}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: The ring has 3 C → cyclopropane.

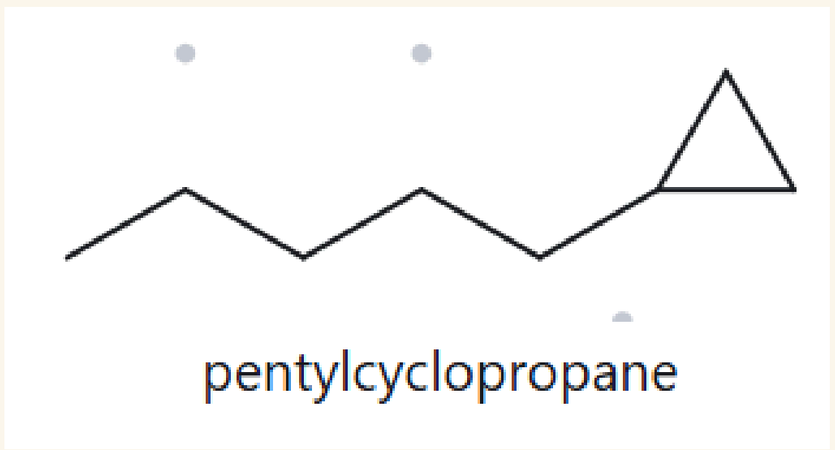
Numbering: Only one substituent is attached to the ring, so a locant is unnecessary.

Substituents: Pentyl attached to the ring.

3. MY NAME

pentylcyclopropane

4. THE ALKANE CHECK



Returned name: pentylcyclopropane

✓ **MATCH**

VERIFICATION STRIP

✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

Part B : NAME → STRUCTURE

Card B01 FOUNDATION

Given Name : 2-methylpropane

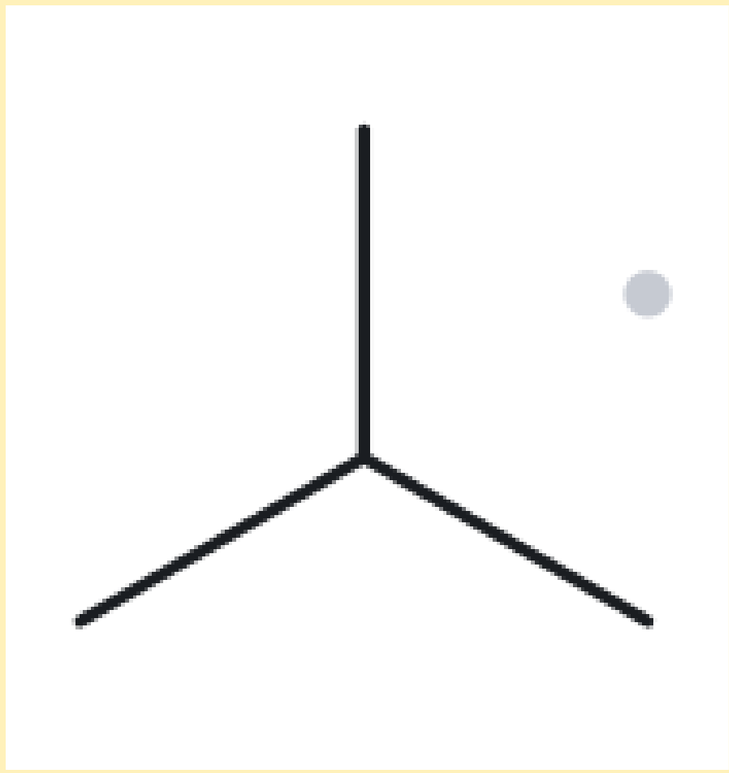
1. NAME DECODED

Parent: Propane → a 3-carbon chain.

Substituents: Methyl on C2.

Expected formula: 3 C; open chain → C_3H_8; add one methyl carbon → C_4H_{10}.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: C_4H_{10}

Open chain: C_nH_{2n+2} = C_4H_{10} ✓

4. BACK CHECK

Returned by The Alkane:

2-methylpropane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

Part B : NAME → STRUCTURE

Card B02 FOUNDATION

Given Name : 3-ethylpentane

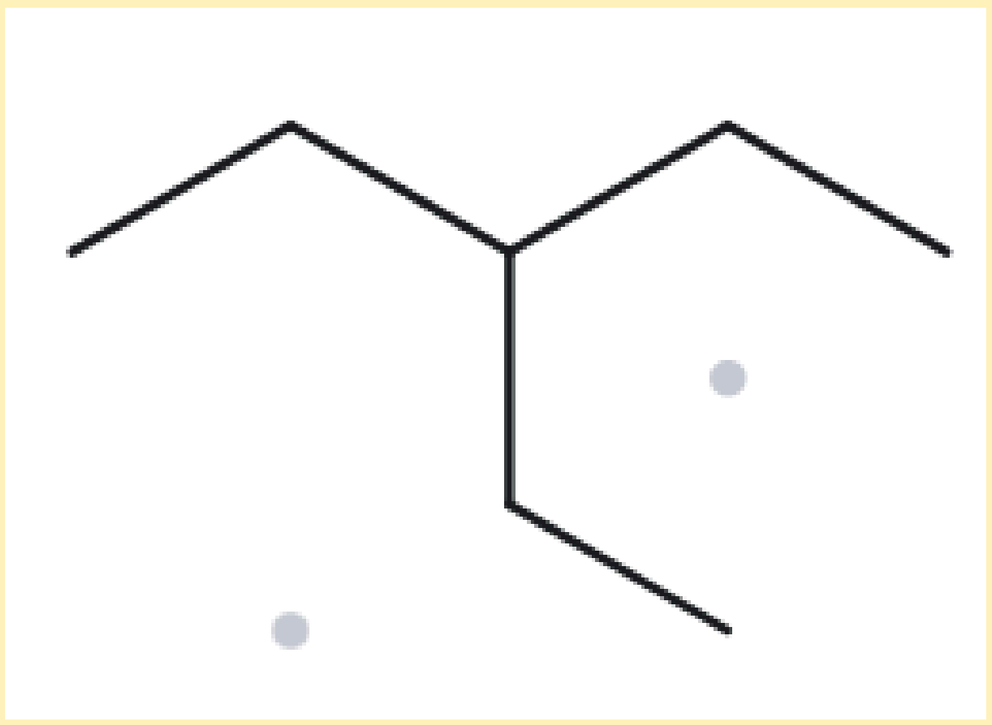
1. NAME DECODED

Parent: Pentane → a 5-carbon chain.

Substituents: Ethyl on C3.

Expected formula: 5 + 2 = 7 C; open chain → C_7H_{16}.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: C_7H_{16}

Open chain: $C_nH_{2n+2} = C_7H_{16}$ ✓

4. BACK CHECK

Returned by The Alkane:

3-ethylpentane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

Part B : NAME → STRUCTURE

Card B03 FOUNDATION

Given Name : 2,2-dimethylbutane

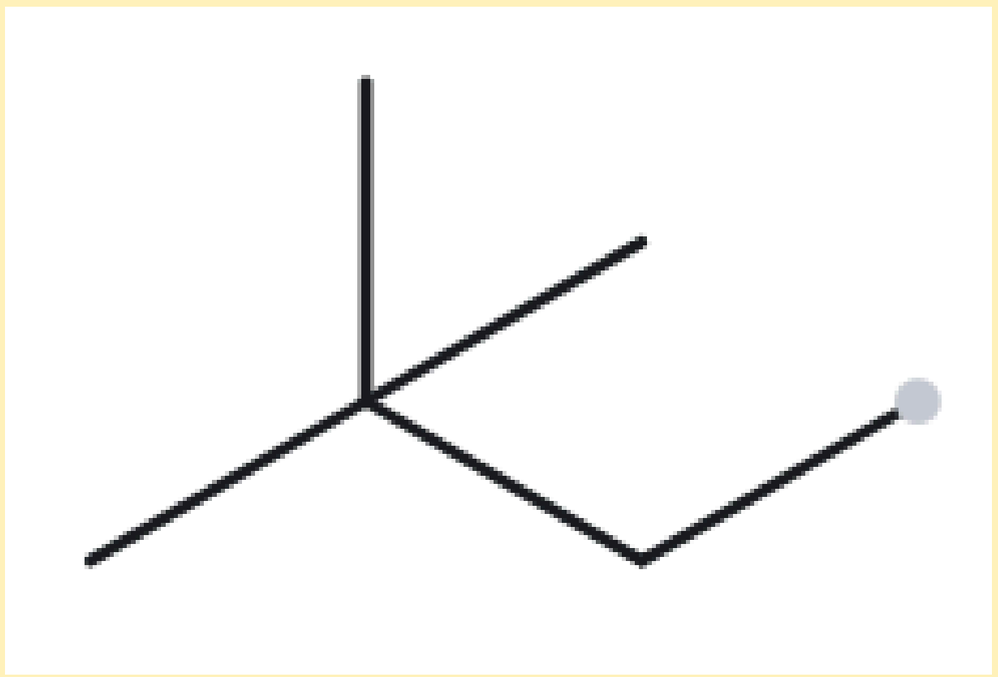
1. NAME DECODED

Parent: butane → a 4-carbon chain.

Substituents: 2 Methyl on C2.

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: C_6H_{14}

Open chain: C_nH_{2n+2} = C_6H_{14} ✓

4. BACK CHECK

Returned by The Alkane:

2,2-dimethylbutane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

Part B : NAME → STRUCTURE

Card B04 DEVELOPING

Given Name : 3-ethyl-2-methylpentane

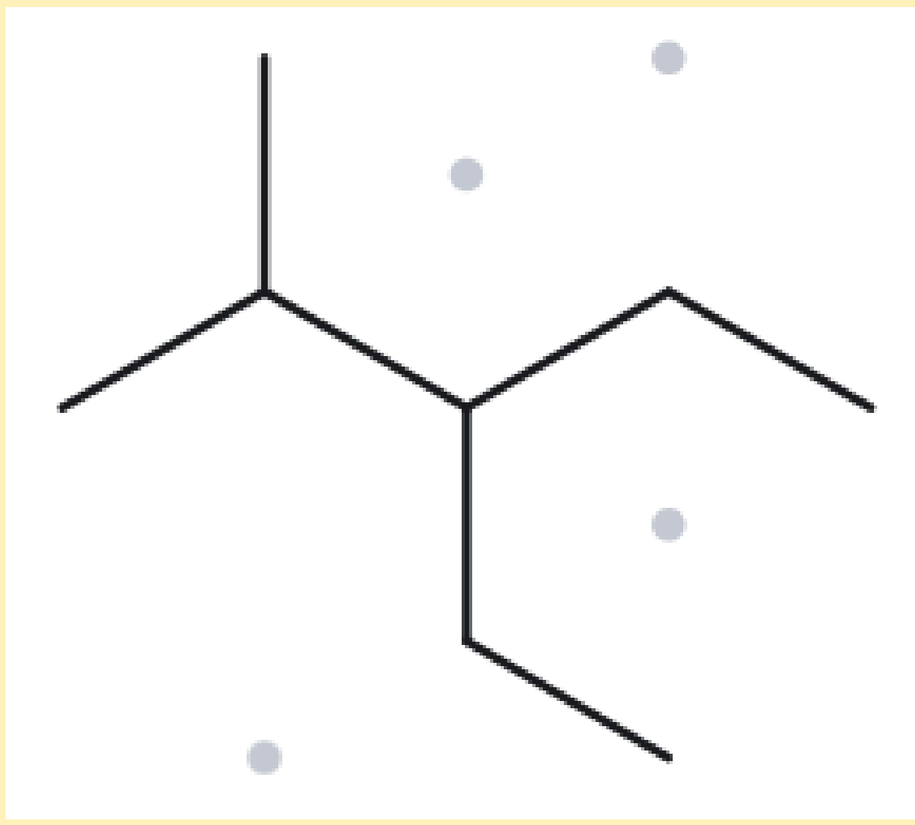
1. NAME DECODED

Parent: Pentane → a 5-carbon chain.

Substituents: Methyl on C2 and ethyl in C3.

Expected formula: 5 + 1 + 2 = 8 C; open chain → C_8H_{18}.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: C_8H_{18}

Open chain: C_nH_{2n+2} = C_8H_{18} ✓

4. BACK CHECK

Returned by The Alkane:

3-ethyl-2-methylpentane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

Part B : NAME → STRUCTURE

Card B05 DEVELOPING

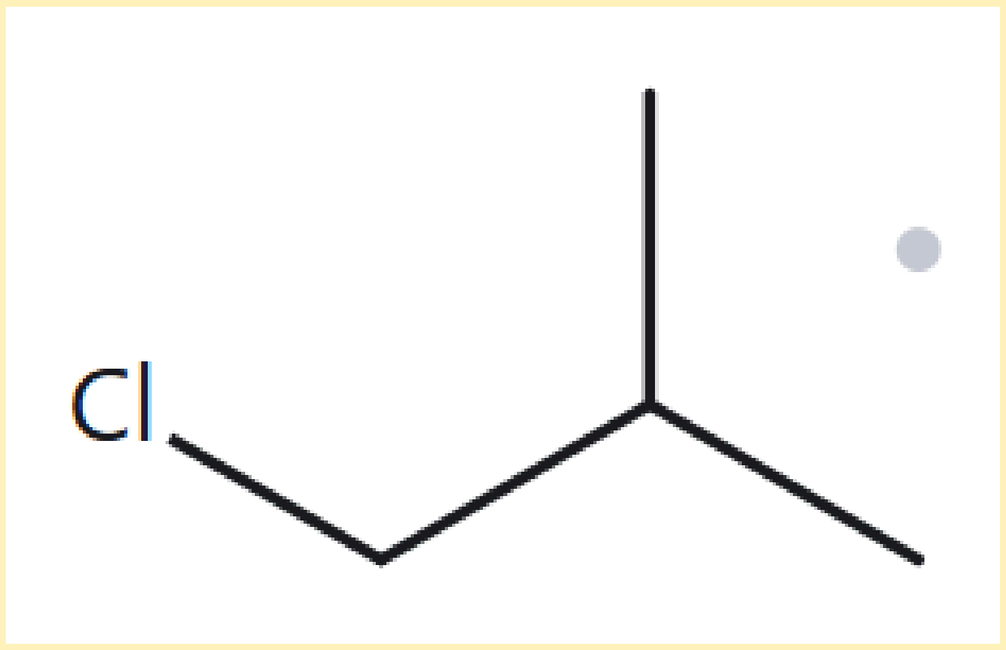
Given Name : 1-chloro-2-methylpropane

1. NAME DECODED

Parent: Propane → a 3-carbon chain.

Substituents: Chloro on C1; methyl on C2.
Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}; one H is replaced by Cl → C_4H_9Cl.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: C_4H_9Cl

Open chain: C_nH_{2n+2} = C_4H_9Cl ✓

4. BACK CHECK

Returned by The Alkane:

1-chloro-2-methylpropane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

Part B : NAME → STRUCTURE

Card B06 DEVELOPING

Given Name : 1,2-dimethylcyclopentane

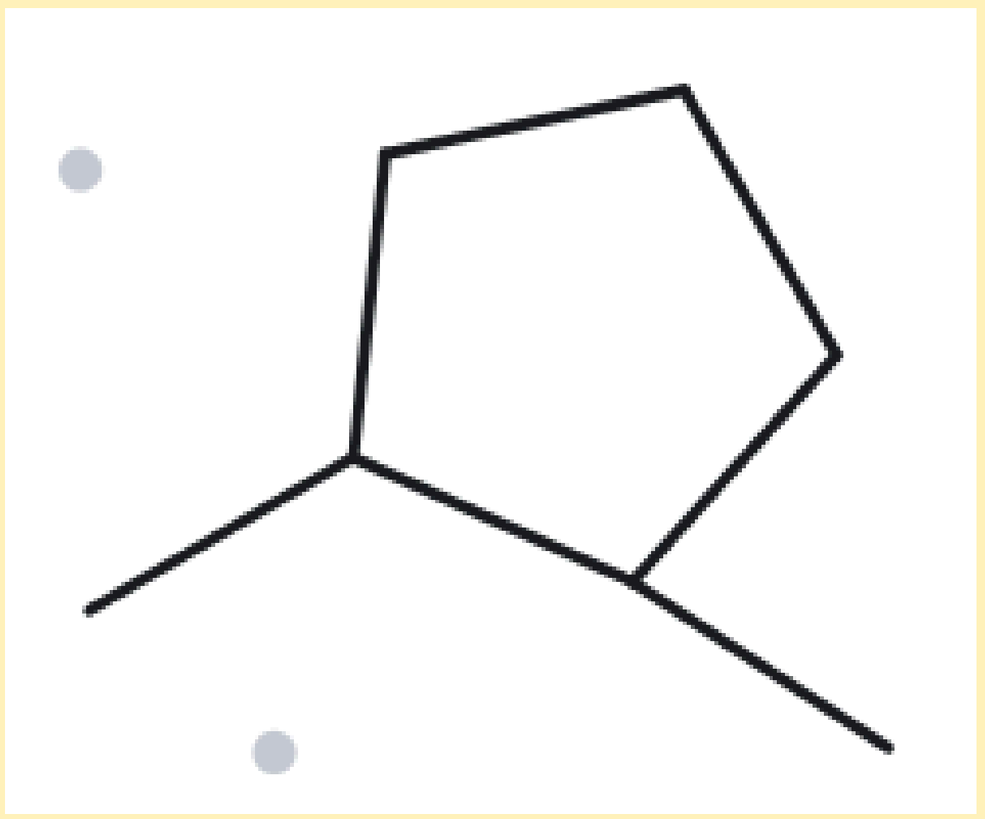
1. NAME DECODED

Parent: Cyclopentane → a 5-carbon ring.

Substituents: Methyl on C1 and methyl on C2.

Expected formula: 5 + 2 = 7 C; one ring → C_7H_{14}.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: C_7H_{14}

One ring: C_nH_{2n} = C_7H_{14} ✓

4. BACK CHECK

Returned by The Alkane:

1,2-dimethylcyclopentane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

Part B : NAME → STRUCTURE

Card B07 DEVELOPING

Given Name : 4-ethyl-3,3-dimethylheptane

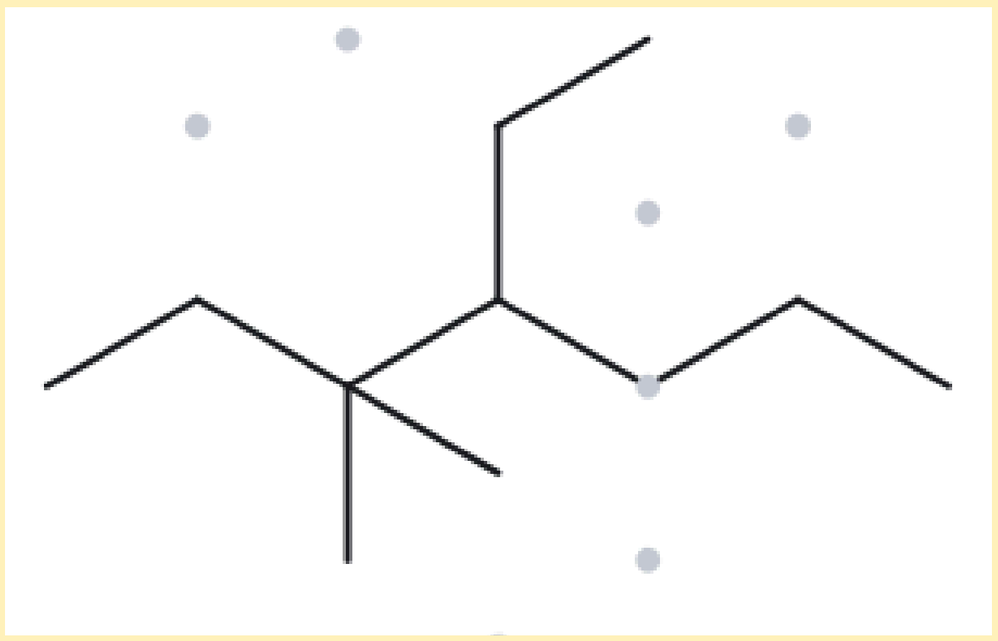
1. NAME DECODED

Parent: Heptane → a 7-carbon chain.

Substituents: Two methyl groups on C3; ethyl on C4.

Expected formula: 7 + 2 + 2 = 11 C; open chain → $C_{11}H_{24}$.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: $C_{11}H_{24}$

Open chain: C_nH_{2n+2} = $C_{11}H_{24}$ ✓

4. BACK CHECK

Returned by The Alkane:

4-ethyl-3,3-dimethylheptane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

Part B : NAME → STRUCTURE

Card B08 CHALLENGE

Given Name : tert-butylcyclohexane

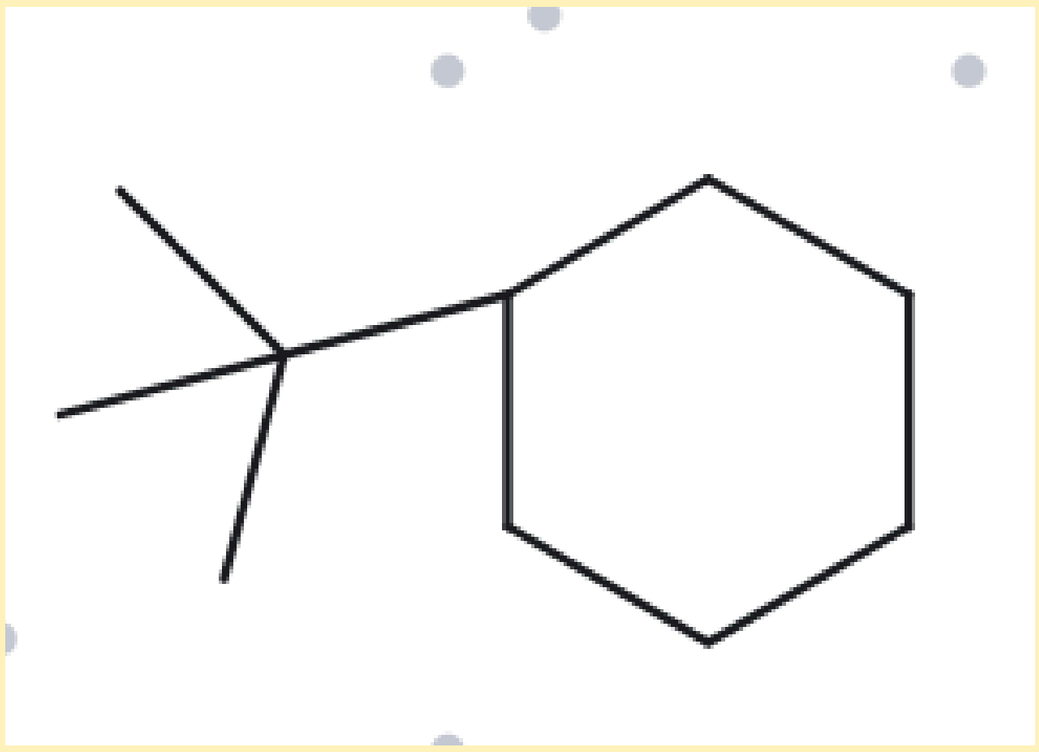
1. NAME DECODED

Parent: Cyclohexane → a 6-carbon ring.

Substituents: Tert-butyl on C1. The tert-butyl group contains 4 carbons.

Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: $C_{10}H_{20}$

One ring: $C_nH_{2n} = C_{10}H_{20}$ ✓

4. BACK CHECK

Returned by The Alkane:

(1,1-dimethylethyl)cyclohexane

Common name: tert-butyl
The Alkane writes: (1,1-dimethylethyl)

✓ Same as the given name

VERIFICATION STRIP

✅ parent length matches ✅ substituent positions match ✅ formula agrees ✅ back-check recorded

Part B : NAME → STRUCTURE

Card B09 CHALLENGE

Given Name : 3-ethyl-1,1- dimethylcyclohexane

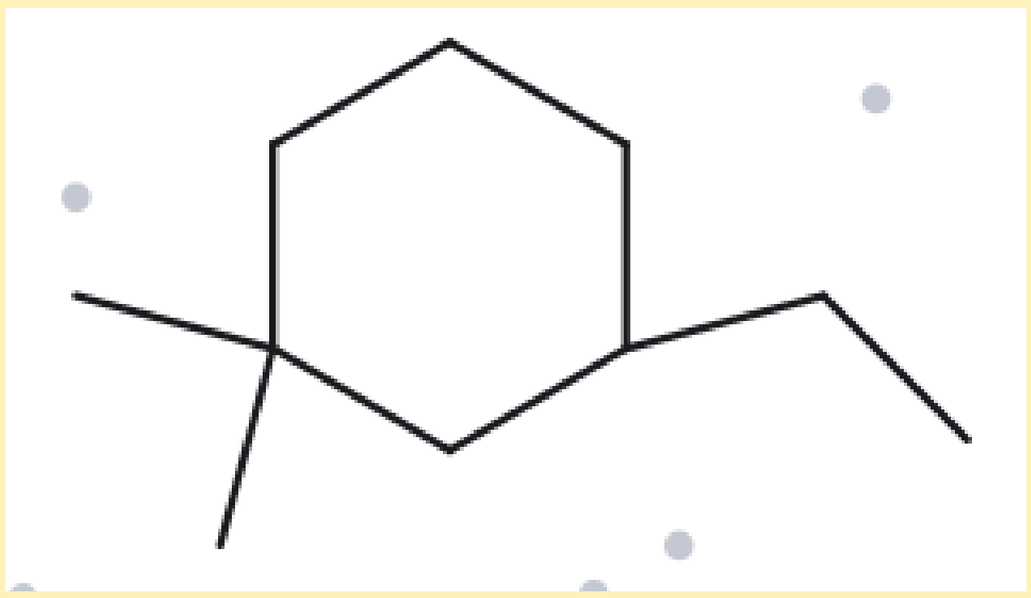
1. NAME DECODED

Parent: Cyclohexane → a 6-carbon ring.

Substituents: Two methyl groups on C1; ethyl on C3.

Expected formula: 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: $C_{10}H_{20}$

One ring: $C_nH_{2n} = C_{10}H_{20}$ ✓

4. BACK CHECK

Returned by The Alkane:

3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

Part B : NAME → STRUCTURE

Card B10 CHALLENGE

Given Name : 2-ethylbutane

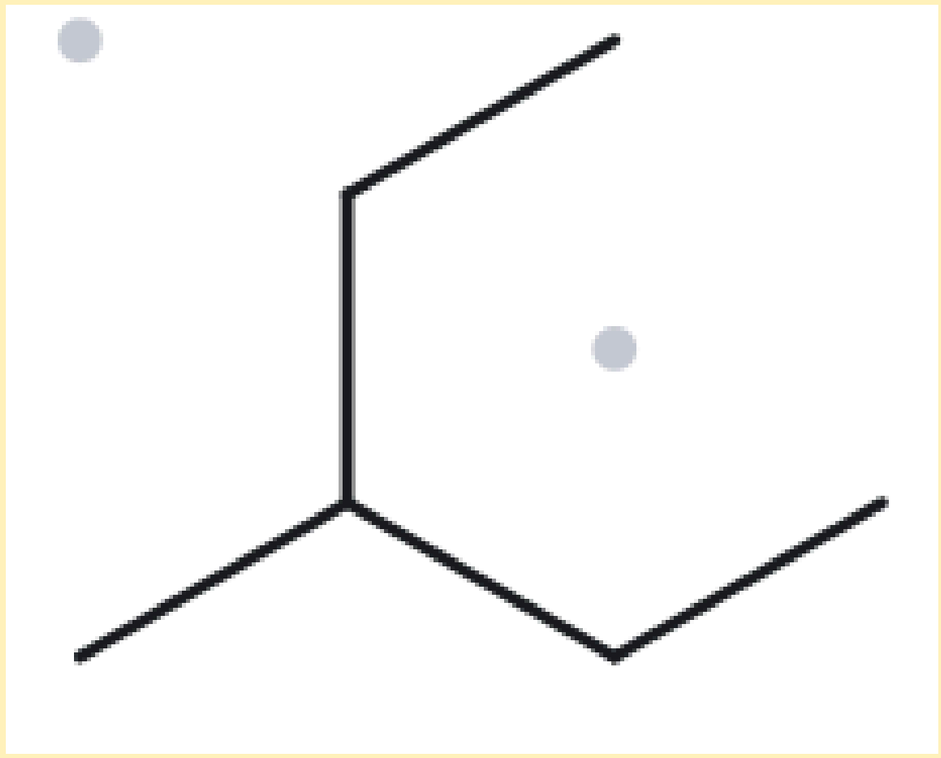
1. NAME DECODED

Parent: Butane → a 4-carbon chain.

Substituents: Ethyl on C2.

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}.

2. BOND-LINE STRUCTURE



3. FORMULA CHECK

Counted formula: C_6H_{14}

Open chain: C_nH_{2n+2} = C_6H_{14} ✓

4. BACK CHECK

Returned by The Alkane:

3-methylpentane

Correct name: 3-methylpentane

Rule broken: The given name does not use the longest continuous carbon chain. The correct parent has 5 carbons (pentane), with a methyl group on C3.

☒ Given name is incorrect

VERIFICATION STRIP

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded